RARE EMBER OFFICIAL SPECIFICATION PRODUCTION CERTIFIED
NEXT.JS 16 • SUPABASE • DELHIVERY AIR

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RARE EMBER™ ARCHITECTURAL & OPERATIONAL MASTER LIFECYCLE

Complete Start-to-End Execution Flow with Every Branch, Failure Mode, State Transition, and Resolution Matrix

Domain: rareember.shop • Atelier: Tirupur Hub (Tamil Nadu) • Carrier: Delhivery Air Express • Gateway: Razorpay Live

> **EXECUTIVE MANDATE & SCOPE**
> This engineering and operations standard details every possible state, branch decision, network event, and edge-case scenario across the entire lifecycle of an order: from storefront catalog discovery, cart calculations, address validation, prepaid/COD payments, atomic order persistence, Tirupur factory queueing, DTF print curing, dual-QA inspection, Delhivery priority air line-haul, Bangalore Hoskote GW inbound sorting, final doorstep handover, and the full 7-day post-delivery returns, replacements, exchanges, RTO recovery, and reverse courier protocols.

1. MASTER END-TO-END SYSTEM ARCHITECTURE

The Rare Ember platform is composed of six loosely coupled, highly resilient architectural planes operating in concert:

Subsystem Plane	Core Technology Stack	Operational Responsibility & Guarantees
1. Storefront & Edge	Next.js 16 (App Router), Turbopack, Tailwind CSS, Vercel Edge Global CDN	Server-side rendering, sub-100ms TTFB globally, dynamic client hydration, CSRF protection, responsive fluid luxury layout.
2. Persistence & Auth	Supabase Postgres 15, Row-Level Security (RLS), Connection Pooler	ACID transactions, atomic stock locks, order history, return claims ledger, rate-limiting counters.
3. Payment Rails	Razorpay Live Gateway, UPI Intent, Webhook HMAC Verification	Prepaid settlement (Cards/UPI/Netbanking), 24h refund reversal API, COD anti-fraud risk evaluation.
4. Atelier & POD Factory	Qikink REST API v2, Tirupur Atelier Automated Queue	240 GSM bio-washed Tirupur combed cotton blanks, 300 DPI high-def DTF curing at 165°C, dual QA inspection.
5. Logistics & Telemetry	Delhivery Air Express, B2B Telemetry Webhooks	Direct airway bill generation, Coimbatore Pudhupalayam GW pickup, priority cargo flight transit, Hoskote GW sorting.
6. Omnichannel Alerts	Resend API, Telegram Founder Alerts, WhatsApp Cloud API	Immediate GST invoice emailing, 1-click WhatsApp concierge tracking, founder Telegram escalations on return claims.

2. THE MASTER 9-STAGE BRANCHING LIFECYCLE

Every customer interaction navigates through a deterministic tree of branching paths. Below is the complete structural decision tree detailing all primary and alternate branches:

[STAGE 1: DISCOVERY & CART] Customer browses rareember.shop catalog ├── Branch 1A (Catalog Selection): Product Detail Page (/product/[id]) | ├── Height & Weight Calculator used → Recommended Size auto-highlighted | ├── Exact Garment Measurements displayed (Chest, Length, Shoulder, Sleeve) | └── Color swatch clicked → Studio daylight 5500K preview rendered ├── Branch 1B (Cart Add & Calculations): | ├── ₹100 express air shipping built-in to retail price (Zero cart surprise) | ├── 40%+ gross margin target verified programmatically | └── LocalStorage synced; Cart Drawer opens with live delivery countdown └── Branch 1C (Abandoned Cart Trigger): └── User drops off → Client session logged; exit-intent preservation active [STAGE 2: CHECKOUT & ADDRESS VALIDATION] Navigation to /checkout ├── 6-Digit PIN Code entered (e.g. 562107) → Auto-lookup executes via postal hub API | ├── PIN Serviceable: Green "Express Air Serviceable" badge displayed (City/State auto-filled) | └── PIN Non-Serviceable (ODA): India Post SpeedPost fallback route flagged ├── Mobile Number entered → Normalized to 10-digit Indian standard (+91) └── Payment Method Branch: ├── Branch 2A: Prepaid Online (Razorpay Live) | ├── Razorpay Modal opens (UPI, PhonePe, GPay, Credit/Debit Cards, NetBanking) | ├── Sub-branch 2A.1 (Payment Success): Signature HMAC verified → Order locked | ├── Sub-branch 2A.2 (Payment Failed / Cancelled): Error banner → Cart preserved for retry | └── Sub-branch 2A.3 (Payment Timeout / Abandoned): Webhook reconciliation check └── Branch 2B: Cash on Delivery (COD) ├── Algorithmic RTO Risk Score calculated (Pincode history + address density) ├── Low Risk → Instant order confirmation └── High/Medium Risk → WhatsApp OTP verification trigger queued [STAGE 3: PERSISTENCE & IMMEDIATE NOTIFICATIONS] ├── Database atomic insert into "orders" table (Supabase Postgres) ├── Automated Telegram alert dispatched to CEO & Atelier desk (notifyTelegramOrder) ├── Resend email sent with formal Tax Invoice PDF attachment ├── Post-Order Experience (/success): | ├── Rare Ember Digital Archival Ownership Pass (RE-XXXX) | ├── Live 5-stage production timeline displayed | └── 2-hour grace window banner (Free 1-tap WhatsApp size change / cancellation) └── Grace Window Branch: ├── Normal Flow: 2 hours elapsed → Order sent to factory production ledger ├── Size Modification: Customer messages WhatsApp → Specs updated in DB └── Cancellation: Customer cancels → Immediate void; Razorpay instant refund [STAGE 4: TIRUPUR ATELIER PRODUCTION & DUAL QA] ├── Factory pull: 240 GSM 100% Super-Combed Bio-Washed Tirupur Cotton blank ├── High-def DTF print applied: 300 DPI master graphic cured at 165°C with dual press ├── Dual-QA 12-point inspection (Chest tolerance, seam tension, print stretch test) | ├── QA Passed: Barcode attached, folded in archival sleeve, packed in tamper-proof mailer | ├── QA Defect Detected: Garment scrapped internally; instant reprint (0 customer impact) | └── Blank Out-of-Stock: Automated alert; customer given option for priority substitute └── Order status updated to "processing" → "shipped" [STAGE 5: COURIER AIR MANIFESTATION & LINE-HAUL] ├── Delhivery Air Express AWB allocated (e.g. 46600611743280) ├── Milestone 1: Manifest uploaded at HQ ├── Milestone 2: Shipment picked up at Coimbatore_Pudhupalayam_GW (Tamil Nadu) ├── Milestone 3: Weight & volumetric dimensions captured at Origin Center ├── Milestone 4: Bag added to priority line-haul trip ├── Milestone 5: Vehicle departed Coimbatore Gateway ├── Milestone 6: Priority domestic cargo flight transit ├── Milestone 7: Trip arrived at Bangalore_Hoskote_GW (Inbound Sorting) └── Logistics Anomaly Branches: ├── Flight Delay / Weather Hold: Automated ETA update on /track ├── Misrouted Hub: Delhivery B2B API re-routing ticket created └── Damaged in Transit: Courier insurance claim filed; duplicate parcel dispatched [STAGE 6: DOORSTEP DELIVERY & OTP HANDOVER] ├── Parcel reaches local delivery depot → Assigned to field rider ├── Checkpoint scan: "Out for Delivery" (OFD) → SMS/WhatsApp sent to customer with OTP ├── Rider reaches doorstep: | ├── Prepaid Handover: OTP entered by rider → Handover complete | └── COD Handover: Exact amount (e.g. ₹548) collected in cash or UPI QR → OTP verified ├── Checkpoint scan: "Shipment Delivered" (T0 = Clock Start for 7-Day Guarantee) ├── Resend email sent: Delivered Tax Invoice + Care & Washing instructions └── Non-Delivery (NDR) Branch: ├── Door locked / Customer unavailable / Cash not ready → Checkpoint: "Undelivered" ├── NDR WhatsApp bot triggers: Customer reschedules delivery for next morning ├── Attempt 2 & 3 executed └── 3 attempts exhausted → Return to Origin (RTO) initiated (Package returns to Tirupur) [STAGE 7: POST-DELIVERY & 7-DAY GUARANTEE WINDOW] ├── Path 7A (Delighted Customer): Garment fits perfectly → Customer joins VIP drops └── Path 7B (Post-Delivery Claim Submitted within 7 Days): └── Form filled on /returns, /track, or WhatsApp (+91 91182 11860) [STAGE 8: RESOLUTION BRANCHES (REPLACEMENT, REFUND, EXCHANGE, RTO)] ├── Branch 8A (Size Mismatch): 10-sec photo check → Free reprint dispatched; customer keeps original ├── Branch 8B (Print Defect): High-res photo → Dual QA approval → 12-hr factory reprint ├── Branch 8C (Transit Damage): Photo of torn bag → Carrier claim filed → Instant reprint ├── Branch 8D (Prepaid Refund): Razorpay API reversal (24-48h back to original bank/UPI) ├── Branch 8E (COD Refund): Payout via instant UPI IMPS (24h) to customer VPA ├── Branch 8F (Store Credit): 110% voucher code generated with lifetime validity └── Branch 8G (Physical Reverse Courier): High-value orders → Delhivery reverse pickup

3. COMPLETE RETURNS, REPLACEMENTS & EXCHANGES SPECIFICATION

Rare Ember rejects conventional, adversarial return policies. Below is the exact operational matrix governing all post-delivery claims:

Claim Reason	Required Customer Proof	Resolution Workflow	Physical Return Required?	Turnaround SLA
Size Mismatch (Fit Exchange)	Photo of collar size tag with measuring tape across chest	**Zero-Return Free Replacement:** Factory pulls requested size (S/M/L/XL/2XL), prints, and dispatches via Delhivery Air. Customer keeps or gifts original.	NO (ZERO RETURN)	24 Hours
Print Defect / Peeling	Clear photo or 5-second video under good lighting showing the flaw	**Priority Factory Reprint:** Production manager recalibrates curing press to 165°C and prints fresh piece in priority slot.	NO (CUSTOMER KEEPS)	12 Hours
Transit Damage (Torn Bag)	Photo of damaged outer Delhivery poly mailer & torn fabric	**Emergency Duplicate Dispatch:** New garment dispatched immediately. Operations files insurance recovery with Delhivery B2B.	NO	6 Hours
Wrong Item Dispatched	Photo of received garment and packing barcode sticker	**Express Correction:** Packing station scan video reviewed. Correct silhouette dispatched via Priority Flight + 15% VIP apology coupon.	NO	12 Hours
Pure Return (Prepaid Refund)	Photo showing unworn garment with tags intact	**Razorpay Direct Reversal:** Automated refund triggered via Razorpay API to original source account (UPI, Card, NetBanking).	YES (HIGH VALUE ONLY)	24–48 Hours
Pure Return (COD Refund)	Photo of unworn garment + Customer UPI ID (VPA)	**Direct UPI IMPS Payout:** Statutory refund sent via instant UPI IMPS to customer's Google Pay/PhonePe ID or Bank NEFT.	YES (HIGH VALUE ONLY)	24 Hours
Store Credit Preference	Standard claim submission on /returns	**110% Bonus Voucher:** Single-use promo coupon issued for Order Total + 10% bonus credit with zero expiry date.	NO	Instant (1 Hour)

4. PHYSICAL REVERSE LOGISTICS & WAREHOUSE INTAKE

For high-value returns (orders > ₹3,000) or suspected return fraud where physical recovery is mandated:

1. **Reverse Waybill Generation:** Delhivery Reverse API (`/api/cmu/create` with `shipment_type: "REVERSE"`) assigns a dedicated reverse tracking ID.
2. **Doorstep QC Inspection (Delhivery Field Agent):**
   - Garment physical presence verified against manifest title.
   - Rare Ember woven atelier neck label or tag confirmed attached.
   - Absence of heavy wear, wash smell, perfume scents, or deliberate cuts verified.

- Customer provides Reverse OTP to courier rider to seal the return bag.

3. **Reverse Line-Haul Routing:**
   - Inbound Pickup Hub: `Bangalore_Hoskote_GW` (Karnataka)
   - Regional Air Hub: Kempegowda International Airport (BLR) $ o$ Coimbatore Airport (CJB)
   - Destination Depot: `Coimbatore_Pudhupalayam_GW` $ o$ Tirupur Atelier Intake QA
4. **Depot Intake Grading & Disposition:**
   - **Grade A (Pristine):** Repackaged in archival glassine pouch, restored to blank inventory.
   - **Grade B (Sample Quality):** Transferred to studio archive for lookbook photography or staff sample sale.
   - **Grade C (Damaged / Unsellable):** Ethically recycled; write-off claimed against courier insurance.

5. FINANCIAL PROOF: ZERO-RETURN VS TRADITIONAL REVERSE COURIER

Mathematical proof demonstrating why Rare Ember's Zero-Return replacement model is financially superior and customer-accretive:

Operational Cost Component	Traditional Reverse Return Model	Rare Ember Zero-Return Replacement Model
Forward Domestic Air Shipping	₹100 (Built into retail)	₹100 (Original order)
Reverse Courier Pickup Fee	₹140 (Standard reverse air)	**₹0 (No reverse booking needed)**
Warehouse Intake & Manual QC	₹45 (Staff inspection labor)	**₹0**
Re-labeling, Re-bagging & Stock Write-down	₹25 (Dead stock buffer)	**₹0**
Fresh Garment Reprint Cost	₹0	₹220 (240 GSM Blank + DTF Print)
Customer Turnaround Time	10 to 14 Days of friction	**24 to 48 Hours**
Customer Lifetime Retention (NPS)	32% (High post-return churn)	**94% (Exceptional brand advocacy)**
Total Direct Operational Cost	**₹310 Net Loss**	**₹220 Total Cost (Saves ₹90/case)**

6. 24 CRITICAL FAILURE MODES & RECOVERY PROTOCOLS

ID	Failure Mode / Edge Case	Systemic Risk	Autonomous & Operational Recovery Procedure
E01	Customer enters invalid/typo Order ID on /returns	Database 404; customer anxiety	Smart resolver checks for partial 8-char prefixes, registered email, or phone number. Displays auto-matched active orders.
E02	Customer submits phone with spaces, dashes, or +91	String mismatch during ownership validation	Normalized server-side using `.replace(/D/g, '')` and matched against the last 10 digits of the recorded shipping phone.
E03	Claim submitted on Day 8 (169 hours post-delivery)	Automated 7-day cutoff rejection	System explains 7-day window. Concierge Bot grants a one-time 24-hour courtesy exception for first-time customers.
E04	Customer files duplicate claim while previous is pending	Double printing / inventory waste	Deduplication filter intercepts request; returns existing `RE-CLM-XXXX` ticket status with live update.
E05	Rate-limit threshold reached on /api/returns	Denial of Service / bot spam	IP rate limiter restricts submissions to 3 per hour per IP. Returns HTTP 429 with polite retry-after header.
E06	Serial return claimer (Abuse / fraud pattern)	Margin erosion across drops	Accounts with >2 lifetime claims flagged in Supabase. System requires physical reverse courier handover before replacement.
E07	Delhivery field rider fails delivery attempt 1 (Customer away)	False RTO trigger	NDR webhook alerts Rare Ember engine. Automated WhatsApp sent to customer with 1-click slot rescheduling.
E08	Customer refuses COD parcel at doorstep due to hesitation	RTO return logistics fee	NDR bot dispatches an instant 10% UPI discount payment link to convert COD into paid online before the courier departs.
E09	Delhivery marks parcel as "Lost in Transit" / Cargo theft	Customer left waiting indefinitely	Telemetry monitor auto-detects lack of scan updates for 72h. Emergency duplicate reprint triggered with priority courier.
E10	Razorpay webhook delayed or dropped by network blip	Paid order stuck in "pending" status	Fallback client verification script polls Razorpay API using `payment_id` to confirm capture and promote order to "paid".
E11	Customer entered incomplete address (Missing house/flat no.)	Rider cannot locate destination	Address sanity regex detects missing structural tokens; prompts customer for WhatsApp landmark clarification before dispatch.
E12	Customer provided invalid UPI VPA handle for COD refund	IMPS payout rejection	Razorpay Fund Account validation pre-checks VPA syntax and bank registration before executing payout; prompts user on error.
E13	Factory stockout on specific color/size blank in Tirupur	Production stall	Atelier inventory sync updates catalog; concierge offers customer upgraded silhouette or guaranteed 48h dispatch date.
E14	Customer moves to a different city before replacement ships	Delivery to wrong address	Replacement workflow allows customer to specify a revised destination address in claim notes before AWB generation.
E15	Customer demands cash from delivery rider during return	Field friction / confrontation	Prominent legal notice displayed on receipts: couriers cannot carry cash. Payouts executed via

	pickup		instant statutory UPI transfer.
E16	Garment damaged by customer bleach / domestic iron burn	Ineligible claim submitted	Polite rejection issued with fabric care guide and a courtesy 25% replacement drop voucher code to preserve goodwill.
E17	Severe weather / cyclone delays air cargo flight to Bangalore	Delayed doorstep ETA	Live radar tracker on /track displays hub weather advisories and dynamically recalibrates estimated delivery date.
E18	Customer requests second replacement after first replacement	Unbounded exchange loops	Policy enforces 1 free replacement per order. Second exchange requires nominal ₹150 domestic courier handling fee.
E19	Bank OTP timeout during Razorpay card checkout	Cart abandoned at payment	Checkout page preserves form inputs, displays "Payment interrupted" alert, and offers 1-tap UPI QR scan alternative.
E20	Tamper-proof poly mailer slashed during airline baggage transfer	Garment soiled / pilfered	Customer submits unboxing photo. Immediate reprint dispatched; corporate claim filed against Delhivery carrier insurance.
E21	Supabase database connection pool exhausted	Order creation failure	Prisma/Supabase connection retry backoff logic (3 retries with jitter); edge fallback logs order to secure SQS queue.
E22	Customer attempts to change size 6 hours after placement	Blank already cured in printer	Atelier desk checks printing station log. If in print queue, rerouted to next batch; if dispatched, handled via /returns.
E23	Resend email bounces due to full customer mailbox	Customer lacks tax invoice	Order tracking page (/track) and invoice page (/invoice) serve printable tax invoice on demand with zero email dependency.
E24	Chargeback dispute filed with credit card issuer	Financial clawback & bank penalties	System compiles full fulfillment package: Proof of Delivery (POD) signature, Delhivery GPS scan, and unboxing log for bank win.

7. TECHNICAL DATA CONTRACTS & STATE MACHINE

The operational state machine in Supabase Postgres enforces rigid legal transitions across all order and claim entities:

```
[LEGAL ORDER STATUS TRANSITIONS] pending ──> paid OR cancelled pending_cod ──> paid OR cancelled paid ──> processing OR cancelled OR refunded processing ──> shipped OR delivered OR cancelled OR refunded shipped ──> delivered OR refunded delivered ──> returned (Terminal state) returned ──> [TERMINAL] cancelled ──> [TERMINAL] refunded ──> [TERMINAL]
```

Supabase Schema: return_claims Table

```
CREATE TABLE public.return_claims (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    claim_id VARCHAR(32) UNIQUE NOT NULL,            -- e.g. 'RE-CLM-7842A1'
    order_id UUID NOT NULL REFERENCES public.orders(id) ON DELETE CASCADE,
    contact VARCHAR(120) NOT NULL,                   -- Email or 10-digit mobile phone
    reason VARCHAR(150) NOT NULL,                    -- Size Mismatch, Defect, Damage, etc.
    preference VARCHAR(100) NOT NULL,                -- Replacement, Refund, Store Credit
    notes TEXT,                                      -- Customer comments / desired size
    status VARCHAR(32) NOT NULL DEFAULT 'pending',   -- pending, approved, reprint_queued,
dispatched, refunded, rejected
    order_total_inr NUMERIC(10, 2) NOT NULL,
    payment_method VARCHAR(20) NOT NULL DEFAULT 'cod',
    replacement_awb VARCHAR(64),                     -- Replacement Delhivery Waybill
    refund_arn VARCHAR(64),                          -- Razorpay ARN or UPI Transaction Reference
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

8. SERVICE LEVEL AGREEMENTS (SLA) & ESCALATIONS

Service Channel	Endpoint / Access Point	Operating Hours	Guaranteed SLA Standard
Automated Returns Portal	rareember.shop/returns	24 Hours / 7 Days / 365 Days	Instant Ticket Generation (< 60 seconds)
Live Radar Tracking Gateway	rareember.shop/track	24 Hours / 7 Days / 365 Days	Real-Time Delhivery Hub Telemetry Sync
VIP WhatsApp Atelier Concierge	+91 91182 11860	9:00 AM – 9:00 PM IST (Mon–Sun)	Under 2 Hours Response Time
Official Support Desk Email	support@rareember.shop	24 Hours Ingestion	Guaranteed Reply under 12 Hours
Founder Escalations	founder@rareember.shop	Direct Executive Escalation	Guaranteed Review under 6 Hours

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